

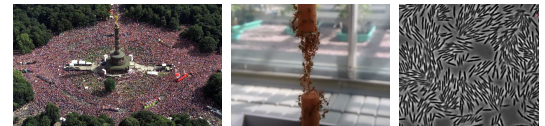
I work on interactive and learning-based algorithms for simulating crowds and modeling the environments they move through. My contributions fall in two connected areas: simulation of multi-agent motion, from pedestrians to ant swarms^{1,2,3,4,5,6,7,8,9,10,11,12,13}, and the analysis and synthesis of 3D scenes, which supply the spaces those agents inhabit^{14,15,16,17,18,19,20,21,22,23,24,25}. These areas are converging on a single goal: realistic, data-grounded virtual environments in which populations of simulated agents can be studied at scale. Such environments allow planners, designers, engineers, and researchers to explore how people move through a space and how changes to it affect outcomes.

My research has been recognized by the *ACM SIGGRAPH Conference on Motion in Games*, where I received the **Best Paper Award**⁹, by the *ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games*, where I received the **Best Student Paper Award**, and by the *ACM SIGGRAPH/Eurographics Symposium on Computer Animation*, where I received the **Best Presentation Award**¹³. In 2018, I was also an internationally selected **finalist** at both the *ACM SIGGRAPH Thesis Fast Forward* and the *ACM SIGGRAPH Asia Doctoral Consortium*. The two thrusts below summarize this work. Thrust I develops crowd dynamics that run in real time at 100,000+ agents and learn behavior from data, and Thrust II develops fast scene synthesis and understanding. I hold this work to three standards: it must be interactive, physically plausible, and, increasingly, validated against observed behavior.

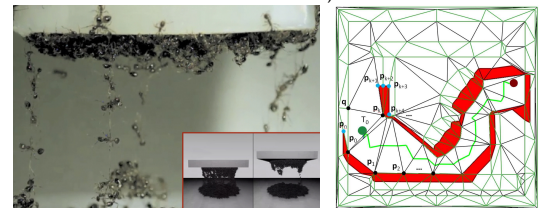
Multi-agent dynamics spans entertainment, pedestrian analysis, urban planning, biology, and robotics. A **crowd** consists of independent, goal-driven agents interacting in a shared space²⁶. Simulating such systems is challenging due to variations in domain, density, and environment, which significantly affect dynamics. For example, methods that specialize in dense crowds smooth individual motion, while sparse-crowd approaches often fail at high densities²⁷.

We proposed a **position-based crowd simulation** method that enables real-time simulation of up to 100K agents⁹. These results are made possible by reframing crowd motion with **dynamics constraints** on agent positions. For example, a constraint for collision avoidance extrapolates each agent's trajectory to detect future collisions. In the event of a collision, agents adjust their trajectory in the direction that **minimizes constraint violation**, with the magnitude of the change depending on the time until an impending collision²⁸. Follow-up work includes investigating when do state-of-the-art methods break²⁹, simulating groups with a position-based **shape matching constraint**⁴, simulating such groups traversing in formation⁵, simulating 3D-capable multi-agent dynamics⁷, and simulating agents with capsules, which better approximate a typical pedestrian profile than a disc³. We also simulated **ant swarms** using position-based dynamics constraints. Ants behave as active matter that morphs and self-repairs, suggesting applications in smart, self-healing materials and multi-agent navigation.

Other work on simulating crowds includes **deep reinforcement learning** and **projective dynamics**. Since "realistic" navigation is difficult to quantify, we train multiple **deep reinforcement learning policies** and infer user preference via **Bayesian inference**^{11,12}. For projective dynamics, formulating crowd motion as a constrained optimization problem enables faster resolution³⁰. We also studied how to improve the



Collective behavior in crowds, ants and bacteria



Ant swarm¹³ simulation (left), and path planning⁵ for crowd formations (right)



Capsules³ better approximate crowd agents than discs (left), resulting in more realistic evacuation simulations (right)



Using crowds to learn how to simulate crowds¹¹

rendering performance of virtual crowds. Users tend to focus on specific elements (e.g., a hero character³¹), so leveraging eye-tracking and foveated information allows full animation in focus regions and simplified updates elsewhere[?].

Outside of simulation related work above, we have also worked on 3D content generation and analysis problems. Since environments in **virtual worlds** are **challenging to construct**, we developed algorithms for generating 3D scenes^{20,17}, using a combination of constrained mathematical optimization and by formulating the synthesis task similarly to text sequence generation, where a scene is represented as a series of tokens. To better understand 3D scene aesthetics for increased realism, we published a method that classifies 3D object style²³. Along this line of work, we also classified real-world room styles^{14,16,15}, a method which is used by Wayfair. Most recently in this thrust, we investigated how rooms and the objects in them affect the perception of clutter and space^{24,25}.

Research Mentoring I have graduated 2 doctoral students, 1 of whom received a faculty position at the DigiPen Institute of Technology. I have also worked with undergraduate and high school students, and published papers with each group.

Next Steps The two thrusts share one idea: complex collective behavior, whether involving people, groups, or arranged objects, can be simulated interactively from simple local rules and then checked against observation. I plan to extend this in three directions. First, I will use crowd and scene simulation for emergency preparedness, modeling how evacuations and interventions unfold in real environments and building virtual training settings in which responders and planners can rehearse decisions that are too costly or dangerous to test in the field. Second, I will carry the multi-agent methods behind our ant aggregation work toward robotics, where groups of simple robots could self-assemble into structures, mimicking the way biological collectives build and repair themselves. Third, I will connect scene synthesis to spatial design, using perceptual measures for generating spaces that are comfortable and usable. The resulting products will be relevant to emergency management, urban planning, transportation, and robotics.

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